

D600 Task 1

B. Demonstrate your understanding of data representation and reporting by writing a paper in which you do the following:

- 1. Create step-by-step instructions to send to a nontechnical executive leader to provide guidance for opening the dashboard and navigating the dashboard using both filters.*

Hello Team,

My name is Aaron and I am a Data Analyst on this project. I have put together a set of step-by-step instructions to help you open and explore the interactive Medical Readmissions Dashboard. I will walk you through opening the dashboard and using the filters to view insights into readmission rates by state and age, patient risk levels, and total charges we are seeing from patients based on their readmission status.

- 1. Locate the Dashboard File*

The file is named: MedicalReadmissionDashboard.twbx

- 2. Open Tableau Public Desktop*

If you do not have Tableau installed, you can download the free version here:

<https://public.tableau.com/en-us/s/download>

- 3. Open the file*

Once you have launched Tableau, look to the upper lefthand corner of the window.

Click File → Open

Then select the MedicalReadmissionDashboard.twbx file from your computer

Now the dashboard should be open and accessible on your screen! From here, I will explain how you can interact with the dashboard to see more specific insights from within the data.

On the right side of the Dashboard, you should see a series of checkboxes. These are the filters and they allow you to see the data through a specific lens.

Filter 1: Complication Risk

This filter allows you to view all insights for patients with a High, Medium, or Low complication risk.

To use, simply check or uncheck the boxes to include or exclude specific risk levels. For example, if you only want to view data for patients with Medium and High complication risks, uncheck the “Low” box and ensure that the Medium and High box are both checked.

Filter 2: State

This filter allows you to break down all data to only patients from specific US states.

To use, First uncheck the box next to “(All)”. This will remove all data from your dashboard. Then, simply find the abbreviation for the state you are interested in, such as “IL” for Illinois, and check that box. This will then show only data from the state you are interested in. If you would like to once again see all data, simply check the “(All)” box once again and the dashboard will automatically update.

If you have additional questions or would like a guided walkthrough, please reach out and let me know!

Thank you,
Aaron Pacheco
Data Analyst

2. Explain how the purpose of your dashboard aligns with the needs outlined in the scenario associated with your chosen dataset.

The purpose of this dashboard is to help executive leaders at a large hospital chain explore trends in readmission data and identify key areas for intervention. This should provide the team with valuable insights to help reduce readmission rates and avoid the CMS penalties for excessive readmissions.

This dashboard is designed to provide valuable insights to each of the different members of the executives teams, with portions aligned to each of their needs.

SVP:

The readmission rate by state and age breakdowns provide the SVP with insights into regional and demographic trends in readmission data that they are most interested in. This allows them to ensure hospitals are complying with readmission regulations and providing high-quality patient care.

EVP:

By providing breakdowns of complication risk and admission type, the dashboard provides the EVP with insights that will assist them in analyzing patterns in patient care, service types, and risk factors that will hopefully lead to data-driven strategies to reduce readmission rates.

CMO:

The dashboard highlights correlations between patient demographics and conditions and the outcomes of the care provided. By highlighting the charges by readmission status and the readmissions by admission type, the CMO is provided with a clearer picture about how provider decisions are affecting readmission rates. This allows the CMO to identify ways care can be improved in order to reduce the risk of readmission.

Overall, this dashboard took a complicated dataset and transformed it into an accessible and interactive tool that will help leaders make decisions about the best possible steps to take to reduce readmission rates, avoid CMS penalties, and improve patient outcomes.

3. Explain **three** different ways you would change your data storytelling approach when presenting to a technical versus nontechnical audience.

When presenting data to different audiences, it is important to keep their backgrounds and goals in mind while preparing your narratives for a presentation. These are the primary factors I would keep in mind and change when talking to a non-technical or technical audience.

1. First would be the depth of the technical details I would provide in my presentation. For a non-technical audience, I would focus on high-level insights and provide non-technical explanations of findings. I would simply provide summaries of findings rather than provide in-depth looks into calculations. Simply stating things such as “We found higher readmission rates among patients admitted through emergency services than other admission types.”

For a technical audience, I would provide more in-depth explanations into things like how the data was prepared, how fields were calculated, and the logic behind my filtering decisions. Technical audiences will have more interest in the methods and techniques of the analysis than simply receiving the summarized findings.

2. The second factor I would change would be the complexity of visualizations. With a non-technical audience, I want to make sure to limit the number of visualizations shown at once and ensure the charts present are relatively simple. Ensuring thorough labeling on visualizations and emphasizing KPIs would also help limit any confusion when presenting the data.

For a technical audience, I would include more detailed or complex visualizations, allowing for deeper exploration. I would also consider adding additional filters which might guide them to further insights not present in a non-technical presentation.

3. The third factor I would focus on shifting when presenting to different audiences would be the language I use and the narrative storyline. For a non-technical presentation, I would be sure to limit technical jargon and speak in everyday language. I would also simplify the narrative structure to a simple “Problem - Evidence - Proposed Solution” style and avoid losing them in the intricacies of the analysis.

With a more technical audience, I could feel more comfortable using statistical terms like “quartiles” and “standard deviations” knowing that my audience wouldn’t get lost in the vocabulary. I would also provide a more analytical storytelling style, going into methodology, assumptions, and accuracy statistics that they might have additional insights into.

4. Identify **two** elements of effective storytelling you could use to present this dashboard, and explain how each element would engage the audience.

1. The first element I would use to present this dashboard would be introducing a narrative flow. This dashboard is designed to follow a narrative pathway from the KPIs at the top of the dashboard down into the supporting visualizations and causes behind these KPIs. This provides a simple flow of “Here are the indicators we are concerned about, now we can take a look at some of the causes.”

This flow can engage audiences because we provide them with the big picture first, then show them what factors led to this big picture. This is similar to how many executives tend to strategize. They see the outcomes, then dig into what the root causes are.

2. The second element I would utilize would be the fact that this dashboard was designed with very specific members of our audience in mind. I’ve explained the specific decisions I made for each executive in Part B2.

Catering each visualization to a specific purpose with an audience member in mind, I should keep each executive engaged with their direct interests and goals and ensure each audience member feels they are personally tied to this analysis.

C. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

No Sources used